Upper Respiratory

1st Generation Antihistamine - Diphenhydramine	
Benadryl Crosses the blood-brain barrier and causes sedation (CNS effects)	
Side Effects & Adverse Effects ● Sedation ● Anticholinergic (dry mouth)	**Education** ● No alcohol or other CNS depressants ● Check combo meds

2nd Generation Antihistamine - Cetirizine	
Zyrtec Less sedation than 1st generation	
Side Effects & Adverse Effects ● Mild sedation	**Education**

Opioid Antitussive - Codeine	
Suppress cough reflex (nonproductive,dry coughs)	
Side Effects & Adverse Effects ● Drowsy, dizzy, constipation ● Respiratory depression, sedation	**Education** ● No alcohol, driving or other CNS depressants ● Naloxone reversal

Non-Opioid Antitussive - Dextromethorphan	
Robitussin - OTC Suppress cough center of brain stem	
Side Effects & Adverse Effects ● Dizzy, drowsy	**Education** ● No SSRIs or MAOIs within 14 days ● Check combo meds

---

Asthma & COPD

Beta2 Agonists

- SNS - NE/E stimulation of beta receptors (Beta2 are mostly lungs and the site of action)
- MOA - stimulate receptors in lungs, relax bronchial smooth muscle, open airway; do not treat underlying inflammation
- Treat airway constriction - COPD, asthma

SABA - Albuterol	
Rescue inhaler Short half-life; 4-6 hour duration	
Side Effects & Adverse Effects • Tachycardia • Tremor • Nervousness	**Education** • CV disease caution • Do not overuse

LABA - Salmeterol	
Maintenance - 12+ hour effects Improve lung function and relax smooth muscle Decrease frequency and severity of COPD or asthma attacks	
Side Effects & Adverse Effects • Tachycardia • Tremor • Nervousness	**Education** • CV disease caution • Often with corticosteroid

Muscarinic Antagonists

- Subset of anticholingerics
- Muscarinic receptors respond to Ach in PNS
- Inhibit Ach at the receptor, blocking bronchodilation and decreasing mucus production

SAMA - Ipratropium Bromide	
Slower onset than albuterol Maintenance med; often used with SABA	
Side Effects & Adverse Effects • Anticholinergic - dry mouth	**Education**

LAMA - Tiotropium Bromide	
Maintenance therapy - consistently open airways and improve over time Used with LABA often - advanced COPD	
Side Effects & Adverse Effects • Anticholinergic - dry mouth	**Education**

Inhaled Corticosteroids - Fluticasone	
Decrease inflammation and inhibit movement of inflammatory cells to site Often combined with LABA in one inhaler	
Side Effects & Adverse Effects	**Education**

• Hoarseness • Oral candidiasis risk • Pneumonia risk in COPD	• Rinse mouth and spit (oral thrush) • Do not stop abruptly

Leukotrine Modifiers - Motelukast	
Block leukotriene receptors to reduce inflammation, improve airflow and prevent tightening Maintenance	
Side Effects & Adverse Effects • Mood/behavior - depression • Black box - suicide • HA	**Education** • Monitor mood, report changes

---

GABA Analogs - Gabapentin	
Neuropathic pain - mimics GABA effects, decreases nerve excitability	
Side Effects & Adverse Effects • Dizzy, sedation, fatigue - falls • R. depression with other CNS depressants	**Education** • Fall prevention • Do not stop abruptly

Dopaminergic Agents - Carbidopa-Levodopa	
Parkinsons - Symptom Management Synergistic - Carbidopa helps more dopamine reach brain Levodopa crosses BBB and is converted to dopamine	
Side Effects & Adverse Effects • Orthostatic hypotension • On/off phenomenon • Dyskinesias • N	**Education** • Protein can decrease absorption; maintain intake consistency • Change position slowly • Do not stop abruptly

Sodium Channel Inhibitors - Phenytoin	
Seizure control, maintenance Small therapeutic window (10-20 mcg/mL) monitor drug serum levels Keeps Na+ channels inactive	
Side Effects & Adverse Effects • Gingival hyperplasia • CNS depression	**Education** • Oral hygiene, dental care • Serum drug level monitoring

GABA Augmenting Agents - Valproic Acid

1st choice for new onset seizures Maintenance	
Side Effects & Adverse Effects ● Hepatotoxicity ● Pancreatitis ● GI, weight gain	**Education** ● Monitor liver function - dark urine, jaundice ● Report abdominal pain, N, V

Benzodiazepines - Lorazepam (for seizures)	
Status epilepticus 1st line rescue	
Side Effects & Adverse Effects ● CNS depression ● R. depression	**Education** ● No alcohol, drowsiness ● Dependence

Methylphenidates - Methylphenidate	
ADHD - increases dopamine and NE activity in CNS Improves attention and focus	
Side Effects & Adverse Effects ● Decreased appetite, weight loss ● Insomnia ● Tachycardia ● HTN ● Dependence - schedule 2	**Education** ● Take in the morning after breakfast ● Monitor weight ● Do not take with MAOIs

Acetylcholinesterase Inhibitors - Donepezil	
Alzheimers - symptom management Increases cognition and memory, improves ADLs Increases Ach	
Side Effects & Adverse Effects ● Cholinergic - N/V/D ● Bradycardia ● Dizzy	**Education** ● Take consistently; weeks for effect ● Report dizziness, fainting and low HR

SSRIs - Fluoxetine	
Depression - elevates mood and decreases irritability	
Side Effects & Adverse Effects ● Sexual dysfunction ● GI ● Black box - suicidal thoughts ● Serotonin syndrome risk	**Education** ● 4-6 weeks for benefit ● Do not stop abruptly ● Report worsening mood ● Do not take with MAOIs

SNRIs - Venlafaxine	
Depression & anxiety	
Side Effects & Adverse Effects ● Sexual dysfunction ● HTN ● Serotonin syndrome risk	**Education** ● Several weeks ● Do not stop abruptly

NDRIs - Bupropion	
Depression, smoking cessation	
Side Effects & Adverse Effects ● Seizure risk ● Insomnia, agitation ● Dry mouth	**Education** ● No seizure disorders ● Weeks to work

Tricyclic Antidepressants - Amitriptyline	
2nd line for treatment resistant depression	
Side Effects & Adverse Effects ● Cardiotoxicity ● Anticholinergic ● Orthostatic hypotension ● Sedation	**Education** ● No CV disease ● Take consistently ● Do not stop abruptly

Monoamine Oxidase Inhibitors (MAOIs) - Selegiline	
Depression not responding to other treatment Prevents breakdown of serotonin, dopamine, NE and tyramine	
Side Effects & Adverse Effects ● Food interactions - tyramine ● HTN crisis ● Drug interactions ● GI, weight gain ● Orthostatic hypotension	**Education** ● Do not combine with SSRI, SNRI, others meds ● Avoid aged cheese, cured meat, fermented foods

Benzodiazepines - Lorazepam (for anxiety)	
Quicker acting; use for 4 months or less Enhance GABA effects	
Side Effects & Adverse Effects ● CNS depression ● R. depression	**Education** ● No alcohol, drowsiness ● Dependence

5-HT1A Partial Agonists - Buspirone	
Generalized anxiety disorder - ongoing maintenance	
Side Effects & Adverse Effects ● Less sedation ● Dizzy, HA, nausea	**Education** ● 2-4 weeks for effect ● NOT PRN ● No MAOIs ● Not for panic attacks

Mood Stabilizers - Lithium	
BPD, acute mania Alters NA ion transport Narrow therapeutic window - drug serum level monitoring	
Side Effects & Adverse Effects ● Tremors - fine hand ● Lithium tox - GI and neuro	**Education** ● Contraindicated kidney disease & dehydration ● NSAID interactions ● Hydrate and consume sodium ● Regular labs

1st Generation (Typical) Antipsychotics - Chlorpromazine	
Schizophrenia	
Side Effects & Adverse Effects ● Movement ● EPS: spasms, contractions, restless, rigid, bradykinesis ● Tardive dyskinesia - irregular, jerky, lip smack, tongue roll ● Sedation ● Orthostatic hypotension	**Education** ● Report abnormal movements ● No alcohol ● Do not stop abruptly

2nd Generation (Atypical) Antipsychotics - Risperidone	
Schizophrenia (1st line)	
Side Effects & Adverse Effects ● Metabolic ● Weight gain, cholesterol, diabetes ● Sedation ● Orthostatic hypotension	**Education** ● Caution in diabetes - monitor weight and glucose